Wings, Lee Bounds and Tail Control

How four models keep their extrapolated wings arbitrage-admissible

Vol-Fitter Technical Notes, No. 09

Vol-Fitter

July 3, 2026

Abstract

A smile model earns its keep in the *wings* — the region beyond the quoted strikes, where the fit becomes pure extrapolation and where mispriced tails leak into var-swaps, risk-reversals and butterflies. This cross-cutting note collects the wing treatment shared by the Vol-Fitter's models. The universal constraint is Lee's moment formula: the asymptotic slope of total implied variance, $\beta = \limsup_{k\to\pm\infty} w(k)/|k|$, cannot exceed 2 and is tied to the moment critical exponents by the explicit map $\beta = \psi(p)$. We show how each model realizes or enforces it — LQD structurally, SVI by a soft cap, Multi-Core SIV by wing-neutral hats — and prove the zero-wing inheritance the SIV design relies on. The finite wing is a separate battle: the asymptotic slope can be admissible while a hat still dents the Durrleman density out where quotes are sparse, which is why the default-on put-wing regularizer samples g *past* the traded range (measured: worst violation $-10.2 \to -0.008$ on the synthetic slice, a $\sim 400\times$ median repair on real illiquid wings, byte-identical on clean slices). A case file replays the backtest finding that motivated it and the ablation verdict that the de-Americanization repair and the penalty are complementary, not redundant ($92 \to 25$ bp with the input repair alone; 749 bp with the output penalty alone; 225 bp with both). The note closes with the arc's centerpiece: the *confinement principle* — which no-arbitrage constraints must be sampled only where data lives, and which must extend into the wings — with its four production instances.

Contents

1 The production problem

Nobody quotes the far wings, yet everything downstream reads them: var-swap replication integrates $1/K^2$-weighted prices into the tails, risk metrics difference the wings, and the graph extrapolator (Note 14) inherits whatever tail the fitted model draws. Wing errors are invisible in the fit RMS — there are no quotes to miss — and show up later as a mispriced tail somewhere else. Two distinct things can go wrong: the *asymptotic* slope can violate Lee's model-free bound (an arbitrage against the existence of any law), and the *finite* wing can lose density positivity even with an admissible slope (a butterfly arbitrage in the extrapolated region).

Invariant

1. Every displayed wing is Lee-admissible: $\beta \leq 2$, structurally where possible, by a soft cap otherwise.
2. Single-slice (butterfly) constraints *extend* into the wings; two-slice (calendar) constraints are *confined* to the traded range (section 5).
3. Every wing guard is strictly additive: on an already-clean slice it is byte-identical to no guard.
4. The tail is a property of the model's extrapolation law, not of the quotes; where the desk needs control (LV deep puts), the slope is an explicit knob or a calibrated variable.

2 The universal constraint: Lee's moment formula

Let $w(k,T) = \sigma_{\mathrm{BS}}^2(k,T)\,T$ be total implied variance at log-moneyness $k = \log(K/F_T)$.

Theorem 1 (Lee moment formula [1])**.** *Define the wing slopes*

$$\beta_R = \limsup_{k\to+\infty} \frac{w(k,T)}{k}, \qquad \beta_L = \limsup_{k\to-\infty} \frac{w(k,T)}{|k|},$$

and let $p_R^* = \sup\{p : \mathbb{E}[S_T^{1+p}] < \infty\}$ *and* $q_L^* = \sup\{q : \mathbb{E}[S_T^{-q}] < \infty\}$ *be the moment critical exponents. Then*

$$\beta_R = \psi(p_R^*), \qquad \beta_L = \psi(q_L^*), \qquad \psi(p) = 2 - 4\left(\sqrt{p^2+p} - p\right). \tag{1}$$

Lemma 1 (Properties of the Lee map)**.** *ψ is a strictly decreasing bijection from $[0,\infty]$ onto $[0,2]$, with $\psi(0) = 2$, $\psi(\infty) = 0$, and explicit inverse $p = \frac{1}{2\beta} + \frac{\beta}{8} - \frac{1}{2}$. In particular $0 \leq \beta \leq 2$ for* every *arbitrage-free smile — the ceiling $\beta = 2$ is model-free.*

Proof. Write $\psi(p) = 2\left(\sqrt{p+1} - \sqrt{p}\right)^2$: the parenthesis is positive and strictly decreasing (its derivative $\frac{1}{2}(1/\sqrt{p+1} - 1/\sqrt{p}) < 0$), with limits 1 at $p = 0$ and 0 at ∞; squaring and doubling gives the range $(0,2]$ plus the endpoint $\psi(\infty) = 0$. Solving $\beta = 2(\sqrt{p+1} - \sqrt{p})^2$ for p gives the stated inverse by elementary algebra. □

The map, and the way LQD reads its slopes off the endpoint scales $A_L = e^{g(0)}$, $A_R = e^{g(1)}$ of Note 01, are a few lines. Appendix C verifies this distilled listing against the production `lee_psi` and `lee_slopes` to 10^{-15} (the production function takes the LQD parameter object and computes the endpoint scales itself):

Listing 1: Lee's ψ and the LQD structural wing slopes (distilled from `models/lqd/basis.py`).

```
def lee_psi(p):                         # beta = psi(p); decreasing, psi(0)=2
    return 2 - 4*(np.sqrt(p*p + p) - p)

def lee_slopes(A_L, A_R):               # exponential tails -> exact exponents
    beta_L = lee_psi(1/A_L)                               # left:  p* = 1/A_L
    beta_R = 2.0 if A_R >= 1 else lee_psi(1/A_R - 1) # right: p* = 1/A_R - 1
    return beta_L, beta_R
```

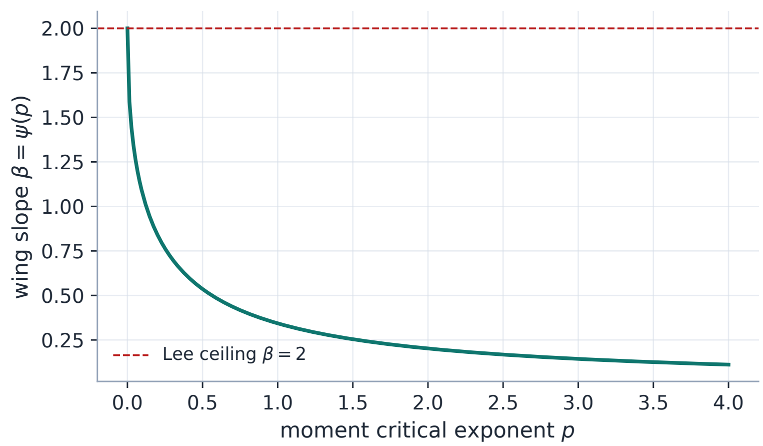

Figure 1: The Lee map $\beta = \psi(p)$: heavier tails (smaller moment exponent p) force steeper variance wings, capped at the model-free ceiling $\beta = 2$.

Heuristic

The wing slope and the tail heaviness are two faces of one coin. A fat right tail (few finite positive moments) *must* show up as a steep right variance wing; a model that draws a flat wing over a fat-tailed density is internally inconsistent and will misprice far-OTM calls. Lee's bound is not a nuisance constraint but a consistency condition between the smile and the law it implies — which is why the LQD model, built from the law outward, satisfies it automatically.

3 Four models, four wing mechanisms

3.1 LQD: Lee-consistent by construction

The LQD model (Note 01) parametrizes the log quantile density with logarithmic endpoints, giving *exponential* return tails with scales A_L, A_R read off the endpoint coefficients. The moment exponents are exact, $p_R^* = 1/A_R - 1$ and $q_L^* = 1/A_L$, and the wing slopes follow from theorem 1:

$$\beta_L = \psi(1/A_L), \qquad \beta_R = \psi(1/A_R - 1), \quad A_R < 1. \tag{2}$$

No penalty enforces this — it is structural. The only wing-related constraint is the hard $A_R < 1$ (finite forward), handled by Note 01's soft barrier. The benchmark fit yields $(\beta_L, \beta_R) = (0.0971, 0.0358)$.

3.2 SVI: linear wings with a soft cap

Raw SVI (Note 02) has exactly linear wings, $w(k) \sim b(1 \mp \rho)|k|$, so $\beta_L = b(1-\rho)$ and $\beta_R = b(1+\rho)$ in closed form ($= (0.1093, 0.0364)$ on the benchmark parameters). Lee's bound is the explicit inequality $b(1+|\rho|) \le 2$, enforced as a soft hinge $\max\big(b(1+|\rho|) - \beta_{\max}, 0\big)$ at weight 10^3 alongside the min-variance penalty (cap `leeSlopeMax` $= 2$). On any admissible smile both penalties are exactly zero; they only fence the optimizer out of the inadmissible region.

3.3 Multi-Core SIV: inherited wings

The Multi-Core SIV slice (Note 03) is a convex base plus up to two *zero-wing* hats.

Proposition 1 (Zero-wing inheritance)**.** *Let $v(z) = v_{base}(z) + \sum_r h_r(z)$ in standardized moneyness z, where each hat h_r and its first two derivatives vanish as $z \to \pm\infty$. Then the asymptotic variance slopes of v equal those of the base, $S_0 \mp 2K_0/\kappa_{P,C}$, for* any *number of hats and any hat parameters; in total-variance k-space, with $k = \sigma_{ref}\sqrt{t}\,z$ and $w = tv$,*

$$\beta_{L,R} = \frac{\sqrt{t}}{\sigma_{ref}}\left|S_0 \mp \frac{2K_0}{\kappa_{P,C}}\right|. \tag{3}$$

Proof. $v'(z) = v'_{\text{base}}(z) + \sum_r h'_r(z) \to v'_{\text{base}}(z)$ as $|z| \to \infty$ since each $h'_r \to 0$; the base is asymptotically linear with the stated slopes. The k-space conversion is the chain rule through $k = \sigma_{\text{ref}}\sqrt{t}\,z$ and $w = tv$. □

On the benchmark, $(\beta_L, \beta_R) = (0.1057, 0.0297)$. Adding cores cannot change the wings — the entire point of the zero-wing design. But the proposition pins only the asymptotic *slope*: a hat can still dent the Durrleman g below zero at *finite* $|z|$, out in the sparsely quoted put wing. That residual failure mode is real, measured, and policed by the put-wing regularizer of section 4.

3.4 Local volatility: controlled linear extrapolation

The piecewise-affine local-vol surface (Note 04) has no closed-form implied-vol wing. Beyond the deepest strike vertex it extrapolates the *local* variance linearly with slope `left_extrap_a` times the first interior cell's slope; the right wing is flat-clamped. The knob mapping is deliberate: the slope is 0 (flat clamp) on the hot path, seeded from `leftWingSlopeMult` $= 1.5$ when the convex-wing hinge is on, and becomes a *free calibration variable* (bounds $[0, 20]$) exactly when a var-swap quote is present to identify the deep-put tail — the one instrument that genuinely observes it (Note 08).

3.5 Side by side

Figure 2 fits the same narrow window with the three parametric models and extrapolates far past it. The wings agree in sign (the equity put skew, $\beta_L > \beta_R$) and order of magnitude but diverge in detail — each tail is a property of *its* extrapolation law, not of the identical quotes. Table 1 collects the analytic slopes.

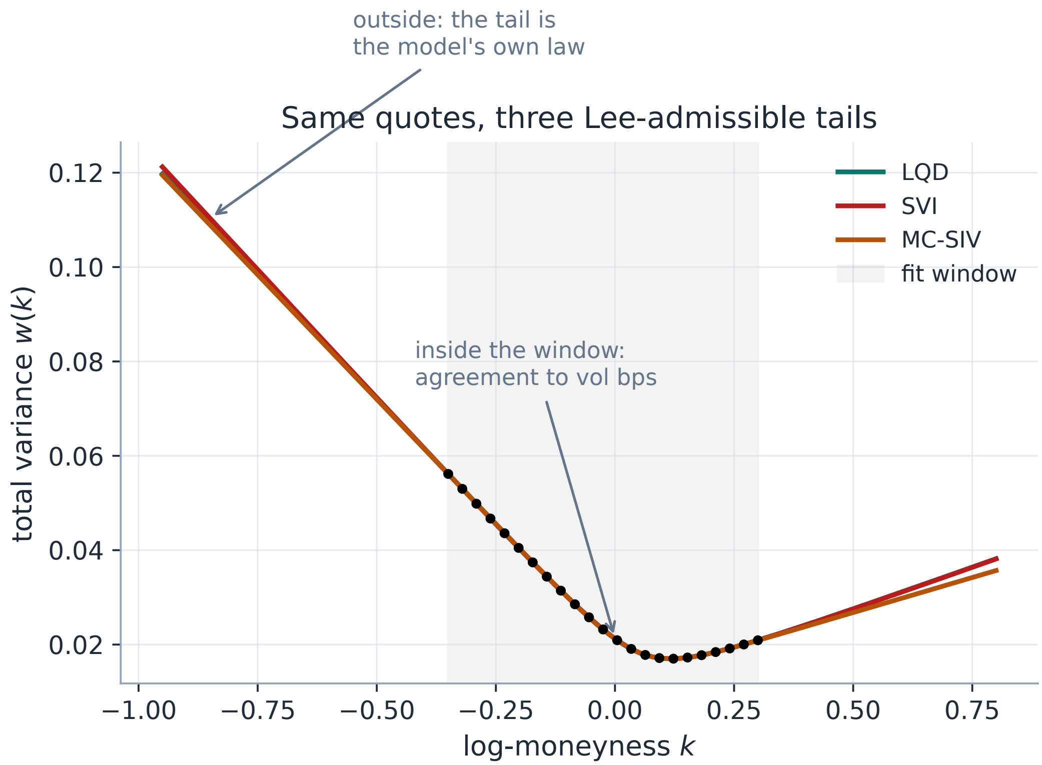

Figure 2: Three models fitted to the same window (shaded) and extrapolated. Inside the window they agree to vol bps; outside, each draws its own Lee-admissible wing. The divergence *is* the model choice.

Table 1: Analytic asymptotic total-variance wing slopes $\beta = \lim w(k)/|k|$ on the benchmark window; all far below the Lee ceiling 2.

Model	β_L (put)	β_R (call)	mechanism
LQD	0.0971	0.0358	structural via $\psi(1/A)$
SVI	0.1093	0.0364	linear $b(1 \mp \rho)$, soft Lee cap
MC-SIV	0.1057	0.0297	base wings, hats neutral (proposition 1)

4 Policing the finite wing: the SIV put-wing regularizer

An admissible slope does not make an admissible wing. The backtest's arb census located the Multi-Core SIV's failures precisely: 64% of butterfly violations sit in the *put* wing (median worst-violation location $z \approx -3.2$), against 4% at the money — hats fitted to sparse illiquid quotes buy in-sample RMS by denting the density where no quote objects. The shipped defence (the R6 fix) has three parts:

1. **A capacity cap.** `nCores` is clamped to ≤ 2 at the schema (and to $(n_{\text{quotes}} - 6)/4$ in the calibrator): most historical violations came from high-core fits chasing noise.

2. **A one-sided Durrleman hinge.** The refine stage carries residual rows $\sqrt{\lambda_{\mathrm{w}}}\,\max(-g(z_m),0)$ on a 49-point grid extending *two standardized-moneyness units past the traded range* on both sides, with the put side weighted 2× (where the census says the risk lives). The weight is $\lambda_{\mathrm{w}} = (\texttt{sivWingPenaltyPct}/100)\cdot 10^3$, default 100%.

3. **A hybrid Jacobian.** The g rows are finite-differenced (49 points, one column at a time) while the quote/ridge/calendar blocks keep their analytic Jacobian — the penalty costs little and perturbs nothing when inactive.

Measured: the synthetic arb slice's worst g went $-10.2 \to -0.008$; on real illiquid wings (EEM) the median worst violation improved $\sim 400\times$ ($-7.86 \to -0.019$) at +79 bp in-sample cost; on clean liquid slices the penalty rows are zero and the fit is byte-identical to 3×10^{-17}. Note that sampling g *past* the quoted range here is correct, not a violation of confinement — see section 5.

> **Case file: the SIV that invented a put wing**
>
> **Setup.** The three-regime backtest sweep (F4): Multi-Core SIV on illiquid single names, where the put wing has a handful of wide quotes.
> **Failure mode.** SIV-2 improved in-sample RMS over SIV-0 — and carried genuine butterfly arbitrage on 76% of arb-prone nodes, almost all in the put wing. The model was buying RMS with a fake weekend wing: a hat parked where no quote could object.
> **Diagnosis.** The analytic (FD-free) arb metric (R2) separated real violations from finite-difference noise: with exact w', w'' per model, the apparent LQD "violations" vanished entirely (28% → 0 — structural positivity confirmed) while SIV-2's survived (76%, genuine). The census then localized them: put wing, $z \approx -3.2$.
> **Fix.** The 2-core cap plus the put-weighted Durrleman hinge above (output side), composing with the de-Americanization wing repair of Note 05 (input side).
> **Verdict (ablation, test-locked).** On the arb-prone illiquid population, with neither fix: worst g -30.4, arbitrage on 100%, in-sample 92 bp. Input repair alone: arbitrage cut $\sim 3\times$ *and* RMS improved to 25 bp — the SIV had been chasing arbitraged de-Am inputs. Output penalty alone: arbitrage essentially eliminated ($g \geq -0.02$) but at 749 bp — brutal, because the model must fight its own corrupted inputs. Both: the penalty's arbitrage removal at 225 bp. *Complementary, not redundant*: the input repair makes the output penalty affordable. On liquid names both are byte-identical no-ops.

5 The confinement principle

Four production incidents — the phantom calendar violations (Note 10's case file), the local-vol convex-wing regression (Note 04), the de-Am global repair revert (Note 05), and the SIV put-wing penalty above — resolve into one rule.

> **Heuristic**
>
> **The confinement principle.** *A constraint that compares two curves must be sampled only where data pins both; a constraint intrinsic to one curve must hold everywhere, wings included.* A two-curve constraint evaluated where both sides are extrapolation can only compare two fictions — any "violation" there is an artefact of the extrapolation laws, and enforcing it lets the fiction corrupt the fit where data *does* live. A single-curve constraint (density positivity, call

> convexity) is a property of the law itself: it has no rival extrapolation to be spuriously violated against, and abandoning it in the wings is exactly how fake tails are born.

Constraint	Curves	Treatment	Where
Calendar floor $w_{\text{far}} \geq w_{\text{near}}$	two	**confined** to traded range	Note 10 (`variance_floor_grid_from`)
LV convex-wing hinge	two[1]	**confined** to the extrapolation tail below the deepest quote	Note 04
SIV Durrleman $g \geq 0$	one	**extended** ± 2 ATM-std past quotes	section 4
De-Am call convexity	one	**extended** to the wings, but band-constrained and ATM-core-fixed	Note 05

The de-Am row shows the principle's refinement rather than an exception: the constraint (convexity) rightly extends into the wings, but the *correction* it licenses is confined — to the quoted bid–ask band and away from the ATM core — because a repair, unlike a constraint, moves prices. Extending a one-curve constraint is safe; extending a repair's *authority* unboundedly is how a 27% put wing became 104% (Note 05).

6 What is original here

The Lee theory is classical; the contribution is the *uniform* treatment. The same slope β is computed three structurally different ways ((2), the SVI closed forms, (3)) and shown to agree on one window, so the model choice is a choice of *how* to extrapolate, never *whether* the extrapolation is admissible. The put-wing regularizer is a census-guided constraint — placed where the measured violations live, weighted by their measured asymmetry, and validated by an ablation that also proved it composes with the input-side repair. And the confinement principle of section 5 — with its two-curve / one-curve criterion — turns four hard-won production incidents into one transferable design rule.

7 Traceability

Table 2: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
LQD structural Lee slopes; underflow-safe	(2)	`models/lqd/basis.py`	`test_lqd_basis.py::test_svi_fit_lee_slopes_match_note`, `::test_lee_slopes_handle_underflowed_endpoint_scales`

[1] The LV wing hinge compares the model's wing against the *quotes'* implied shape when sampled inside the quoted range — the regression of Note 04 — so it behaves as a two-curve constraint and is confined; in the pure extrapolation tail it degenerates to a one-curve convexity preference, which is where it is allowed to act.

Claim	Object	Code anchor	Test anchor
SVI Lee cap enforced	section 3	`models/svi_jw/calibrate.py`	`test_svi_calibrate.py::test_respects_lee_wing_bound`
SIV hats are zero-wing	proposition 1	`models/sigmoid/sigmoid.py`, `kernels.py`	`test_sigmoid_model.py`
2-core cap	section 4	`api/schemas.py`, `models/sigmoid/calibrate.py`	`test_siv_wing_penalty.py::test_cores_clamped_to_two`
Put-wing hinge removes wing arb; byte-identical on clean slices	section 4	`models/sigmoid/calibrate.py`	`test_siv_wing_penalty.py::test_penalty_removes_wing_arb`, `::test_penalty_byte_identical_on_arb_free_slice`
Analytic FD-free arb metric	section 4	`backend/backtest/dispatch.py`, `models/sigmoid/kernels.py`	`test_backtest_arb.py`
R3×R6 ablation: complementary	case file	`backend/backtest/ablation_arb.py`	`test_ablation_arb.py`
De-Am wing repair (input side)	section 5	`calib/convex_deam.py`	`test_convex_deam.py`
Calendar floor confined (two-curve)	section 5	`calib/calendar.py`	`test_overlay_calendar.py::test_wide_grid_breaks_svi_but_data_grid_does_not`
LV convex wing confined to tail	section 5	`api/affine_fit.py`	`test_affine_grid_design.py::test_convex_wing_confined_to_quoted_extrapolation`

A Hyperparameter atlas

Table 3: Wing-related knobs (cross-model).

Knob	Default	Role
Surfaced		
`sivWingPenaltyPct`	100	Put-wing Durrleman hinge strength (% of the 10^3 base weight); 0 disables.
`nCores`	2 (max 2)	SIV core count, schema-clamped (section 4).
`leeSlopeMax`	2.0	SVI Lee cap $b(1+\lvert\rho\rvert) \leq \beta_{\max}$.
`sviPenaltyWeight`	10^3	Weight of the SVI Lee / min-variance penalties.
`leftWingSlopeMult`	1.5	LV left extrapolation slope seed; free variable under a var-swap (Note 04).
`lvVolCapMult`	3.0	LV local-vol cap multiple (Note 04).
Hidden		

Knob	Default	Role
_WING_PAD	2.0	The g-grid extends ± 2 standardized units past the traded range.
_WING_GRID	49	Points on the g-grid.
_WING_PUT_FACTOR	2.0	Extra weight on the put side (the census asymmetry).
EPS_AR	10^{-6}	LQD $A_R < 1$ buffer; the only LQD wing constraint (Note 01).
calendar floor grid	traded range	Confined per the principle of section 5 (Note 10).

B Performance notes

Wing handling is essentially free: the LQD slopes are a by-product of the endpoint scales; SVI/SIV slopes are closed forms of the fitted parameters; the LV wing is part of the surface solve. The put-wing hinge adds 49 residual rows in the refine stage only, with a hybrid Jacobian (analytic blocks untouched, g rows by finite differences); on a clean slice the rows are zero and the optimum is byte-identical. The analytic arb metric (R2) replaced a double `np.gradient` on 201 points with closed-form w', w'' per model — more accurate *and* cheaper. Confinement (section 5) reduces work by not over-constraining the extrapolation region.

C Reference implementation

The listing of section 2 is verified against `models/lqd/basis.py`: `lee_psi` is character-identical, and the distilled `lee_slopes` (taking the endpoint scales directly) agrees with the production function (which takes the LQD parameter vector and computes the endpoint scales itself) to 10^{-15} across the benchmark fits, including the underflow guards at extreme scales.

References

[1] R. W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004.

[2] S. Benaim and P. Friz. Regular variation and smile asymptotics. *Mathematical Finance*, 19(1):1–12, 2009.

[3] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.